Chapter 8 — Process Death & State Restoration

> **Part I — Core Infrastructure** · Complexity: **Standard** (≥25 interview questions)
>
> **Prerequisites:** Ch 3 (SSOT/Room — the durable store), Ch 4 (DataStore — durable config), Ch 6 (UI state & ViewModel). This chapter completes Part I by making screens survive **process death**, not just rotation. Applied in Ch 9 (restore query), Ch 38 (cart), Ch 75 (checkout draft).
>
> **Reference surface in StreamCart:** restoring the search query, a half-filled checkout form, and scroll/selection after the OS kills the app.

8.1 Problem Definition and Real-World Context

Android can **kill your app's process at any time** when it's in the background — to reclaim memory for the foreground app. When the user returns, the system recreates the Activity/task, but **all in-memory state is gone**: ViewModel fields, composable state, in-flight coroutines. If you only handle *rotation* (which `ViewModel` survives) and forget *process death*, users lose their work:

- **Lost input.** A user types a long review or fills a checkout form, gets a call, returns ten minutes later — the form is blank. This is a top complaint in reviews and a conversion killer in commerce/fintech.
- **Wrong screen on return.** Without saved navigation/selection, the app reopens at the start instead of where the user was.
- **"Works on my machine."** Rotation testing passes; process death only shows up under memory pressure, so the bug reaches production.

The distinction is critical: **`ViewModel` survives configuration changes but NOT process death.** Surviving process death requires explicitly saving a small amount of state via **`SavedStateHandle`** (backed by the saved-instance-state `Bundle`) and recovering the rest from the **SSOT** (Room/DataStore, Ch 3/4). Deciding *what* to save where is the architectural work.

Three practical use cases

1. **E-commerce checkout (StreamCart).** A half-filled shipping/payment form and the current step are restored so the user resumes exactly where they left off.
2. **Social compose (Social).** A draft post/comment survives process death (ideally persisted to Room for long drafts — Ch 37), so nothing is lost.
3. **OTT browse (Watch).** The selected category, search query, and scroll position are restored so returning feels seamless.

Why this is architectural

It defines **what state is transient vs durable**, **where each kind lives** (Bundle via `SavedStateHandle` vs Room/DataStore vs re-fetch), and **how the app reconstructs itself** after the OS reclaims it — a system-level

contract, not a per-screen afterthought.

8.2 Requirements Engineering

Functional requirements

ID	Requirement
F1	Key UI input (query, form fields, selection, step) survives process death.
F2	Durable data is recovered from the SSOT (Room/DataStore), not the Bundle.
F3	Restoration is automatic on recreation — no user action.
F4	Saved state is small (Bundle has a hard size limit).
F5	In-flight operations are re-derived or resumed safely, not duplicated.
F6	Distinguish first launch from a restored launch.

Non-functional requirements

Attribute	Policy
Reliability	No lost user input; correct screen on return.
Performance	Save only tiny state; restore without blocking startup.
Correctness	No duplicated side effects on restore (idempotency, Ch 15).
Memory	Bundle ≪ ~1 MB (TransactionTooLarge risk); keep it bytes/KB.
Security	No secrets/PII in the Bundle (written to disk unencrypted).

Scope

- **In scope:** `SavedStateHandle`, `rememberSaveable`, what to save vs re-derive, Bundle limits, restored-vs-fresh detection, interplay with SSOT, re-deriving in-flight work.
- **Out of scope:** durable draft persistence to Room (Ch 37), background work continuation (WorkManager, Ch 30), navigation state framework (Ch 53), auth session restore (Ch 22).

Assumptions & constraints

- `@HiltViewModel` + `SavedStateHandle`; Compose `rememberSaveable`. Test process death via *Don't keep activities* / `adb shell am kill` / background + low memory. `minSdk 24`.

8.3 System Design Fundamentals

- **Process death vs config change.** Config change (rotation) recreates the Activity but keeps the `ViewModel`. Process death destroys *everything*; the system later recreates the task from saved instance state.

- **`SavedStateHandle`.** A key-value map, injected into ViewModels, backed by the saved-instance-state `Bundle`. Values put here survive process death and are restored into the new ViewModel. It also exposes `getStateFlow()` for reactive, saved state.
- **The Bundle is tiny and serialized to disk.** Only small, Parcelable/primitive state belongs here (ids, query text, form fields, scroll key) — never lists, bitmaps, or secrets.
- **SSOT recovers the rest.** Large/durable data isn't saved in the Bundle; it's re-read from Room/DataStore (Ch 3/4) or re-fetched. The Bundle holds *just enough* to re-derive the UI.
- **`rememberSaveable`.** The Compose counterpart for purely-UI state (expanded sections, text field with cursor) via the `SaveableStateRegistry`.
- **Idempotent restore.** Re-deriving must not replay side effects (don't re-submit an order because a "paying" flag restored); use idempotency (Ch 15) and derive from authoritative state.

Core insight: **save the minimal key(s) needed to reconstruct the screen in `SavedStateHandle`, and rebuild everything else from the SSOT.** The Bundle is a pointer, not a cache.

8.4 High-Level Architecture


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Components & ownership

- **`SavedStateHandle`** holds the minimal transient keys (query, form fields, current step) — survives process death.
- **SSOT (Room/DataStore)** holds durable data, recovered on restore.
- **ViewModel** derives `UiState` from both; Compose `rememberSaveable` handles purely-visual state (expanded/collapsed, cursor).

Direction: Bundle → `SavedStateHandle` → ViewModel; SSOT → ViewModel; both → `UiState` → UI.

8.5 Detailed Component Design


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What	Where to save	Why
Search query, form fields, current step, selected id	`SavedStateHandle`	small, transient, must survive process death
Expanded/collapsed, text cursor, tab	`rememberSaveable`	pure UI, per-composable
Catalog, cart, history	Room (SSOT)	durable, large, re-read on restore
Theme, session metadata, onboarding flag	DataStore	durable config
Long draft (e.g., review)	Room (+ debounce save)	too big/important for Bundle (Ch 37)
Tokens/secrets	Keystore (never Bundle)	security

8.6 End-to-End Data Flow


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8.7 Complete Kotlin Implementation

Package: `com.streamcart.checkout`.

8.7.1 Saving/restoring via `SavedStateHandle`

```
@Parcelize
data class CheckoutForm(
    val name: String = "",
    val address: String = "",
    val city: String = "",
) : Parcelable

@HiltViewModel
class CheckoutViewModel @Inject constructor(
    private val savedState: SavedStateHandle,
    observeCart: ObserveCartUseCase,
    private val placeOrder: PlaceOrderUseCase,
) : ViewModel() {

    // Reactive, process-death-surviving state.
    val step: StateFlow<Int> = savedState.getStateFlow(KEY_STEP, 1)
    val form: StateFlow<CheckoutForm> = savedState.getStateFlow(KEY_FORM, CheckoutForm())

    // Durable data recovered from the SSOT (not the Bundle).
    val cart: StateFlow<CartSummary?> = observeCart()
        .map { it.dataOrNull()?.summary }
        .stateIn(viewModelScope, SharingStarted.WhileSubscribed(5_000), null)

    fun onFieldChange(update: (CheckoutForm) -> CheckoutForm) {
        savedState[KEY_FORM] = update(form.value)     // persisted into the Bundle
    }
    fun goToStep(s: Int) { savedState[KEY_STEP] = s }

    fun submit() = viewModelScope.launch {
        // Idempotency key prevents a duplicate order if we restore mid-submit (Ch 15).
        val key = savedState[KEY_IDEMPOTENCY] ?: UUID.randomUUID().toString().also {
            savedState[KEY_IDEMPOTENCY] = it
        }
        when (placeOrder(form.value, idempotencyKey = key)) { /* ... */ }
    }

    private companion object {
        const val KEY_STEP = "checkout_step"
        const val KEY_FORM = "checkout_form"
        const val KEY_IDEMPOTENCY = "checkout_idem_key"
    }
}
```

8.7.2 Compose purely-UI state with `rememberSaveable`

```
@Composable
fun CheckoutScreen(viewModel: CheckoutViewModel = hiltViewModel()) {
    val step by viewModel.step.collectAsStateWithLifecycle()
    val form by viewModel.form.collectAsStateWithLifecycle()

    // Pure-UI state that should survive process death but isn't business state.
    var expandedSummary by rememberSaveable { mutableStateOf(false) }
    val scrollState = rememberScrollState()   // position restored via Saver

    // ... text fields bound to `form`, updating via viewModel.onFieldChange { it.copy(...) }
}
```

8.7.3 Custom `Saver` for a non-Parcelable UI type

```
val FilterSaver: Saver<Filter, List<String>> = Saver(
    save = { listOf(it.category, it.sort) },
    restore = { Filter(category = it[0], sort = it[1]) },
)
// usage: var filter by rememberSaveable(stateSaver = FilterSaver) { mutableStateOf(Filter()) }
```

8.7.4 Detecting restored vs fresh launch

```
val isRestored: Boolean = savedState.contains(KEY_STEP)   // key present ⇒ recreated
```

8.7.5 Module structure

```
:feature:checkout/presentation  CheckoutViewModel (SavedStateHandle), CheckoutForm(@Parcelize)
:feature:checkout/ui            CheckoutScreen (rememberSaveable, Savers)
:data:cart                      ObserveCartUseCase (SSOT recovery, Ch 3)
```

8.8 Concurrency and Race Conditions

Scenario	Risk	Handling
Process death mid-submit	Duplicate order on resume	Persist idempotency key in `SavedStateHandle`; backend dedupes (Ch 15)
Restore triggers refetch + stale write	Overwrite newer data	Read SSOT (authoritative) on restore; don't replay the old network write
`getStateFlow` collected before restore	Default then real value	Default value provided; restored value emitted; use `distinctUntilChanged`
Large object into Bundle	`TransactionTooLargeException`	Save ids/small fields; keep big data in Room
In-flight coroutine lost on death	Operation never completes	Move durable work to WorkManager (Ch 30); re-derive UI from state

- **Idempotency is the key race-safety tool:** a persisted idempotency key means a resumed submit after process death won't double-charge/double-order.
- **Re-derive, don't replay:** on restore, rebuild the UI from authoritative state (SSOT) rather than re-running the side effect that was interrupted.

8.9 Failure Scenarios and Recovery

Failure	Detection	Response	UX
Background process death	recreate with `SavedStateHandle`	restore keys + read SSOT	seamless resume
Bundle too large	`TransactionTooLargeException`	move big state to Room; save ids only	no crash
Secret saved in Bundle	security review	never save secrets; use Keystore	no leak
Mid-operation death	idempotency key present	safe resume; backend dedupe	no duplicate
Draft lost (too big for Bundle)	—	persist drafts to Room with debounce (Ch 37)	draft survives
Restored state inconsistent with SSOT	compare keys vs data	prefer SSOT; reconcile	correct data

8.9 Failure Scenarios and Recovery

> **Test it or it's broken:** enable *Developer options → Don't keep activities*, or `adb shell am kill com.streamcart`, to force process death. Rotation testing does **not** exercise this path.

8.10 Scaling the Implementation

- **Level 1.** Save a single value (query/selected id) in `SavedStateHandle`.
- **Level 2.** Save form fields/step; recover durable data from the SSOT; `rememberSaveable` for UI bits.
- **Level 3 — production.** Idempotency keys for in-flight mutations; `@Parcelize` state; restored-vs-fresh detection; long drafts persisted to Room (Ch 37).
- **Level 4 — large.** A consistent "restorable state" contract across features; navigation/back-stack restoration (Ch 53); scroll/selection restoration in Paging (Ch 13).
- **Level 5 — enterprise.** Formalized session/journey restoration spanning multiple screens; crash+restore analytics; policy on what each feature persists where (Bundle vs Room vs DataStore).

Avoid overengineering: don't persist everything — most durable data belongs in the SSOT already; `SavedStateHandle` holds only the few transient keys needed to re-derive the current view.

8.11 Alternative Approaches and Trade-offs

State kind	Store	Survives process death?	Notes
Transient UI input (query, form, step)	`SavedStateHandle`	✅	small, Parcelable/primitive
Pure-UI composable state	`rememberSaveable`	✅	per-composable; Saver for custom types
In-memory only	ViewModel field / `remember`	❌ (survives rotation only)	ephemeral, recomputable
Durable app data	Room (Ch 3)	✅	large/relational; authoritative
Durable config/session meta	DataStore (Ch 4)	✅	key-value
Long/important drafts	Room (+ debounce)	✅	too big for Bundle (Ch 37)
Secrets	Keystore (Ch 20)	✅ (encrypted)	never Bundle

`SavedStateHandle` vs Room for restoration: use the Bundle for tiny transient UI keys that only matter to the current journey; use Room when the data is large, relational, must be queryable, or must persist beyond a single task (e.g., a draft you want available on a new device after sync). Many screens use both: Room for data, `SavedStateHandle` for "where in the flow am I."

8.12 Security Considerations

- **The saved-instance `Bundle` is written to disk unencrypted** (and historically readable in some OS versions/backups). **Never store tokens, passwords, card numbers, or PII** in `SavedStateHandle` / `rememberSaveable`. Save a reference (id) and resolve the sensitive value from Keystore/encrypted storage (Ch 20).

- **Drafts with PII** (addresses, messages): if persisted to Room, encrypt sensitive tables (Ch 24) and exclude from auto-backup (Ch 37).
- **Idempotency keys** are not secrets but shouldn't be guessable; a UUID is fine.
- Client vs backend: restored client state is a convenience; the backend remains authoritative and must re-validate on any resumed privileged action.

8.13 Performance and Resource Optimization

- **Keep the Bundle tiny.** `onSaveInstanceState` runs on the main thread and the Bundle crosses a Binder transaction (~1 MB limit shared per transaction); large saves cause jank or `TransactionTooLargeException`.
- **Don't block startup on restore.** Reading the SSOT on restore is async (`Flow`); show restored UI keys immediately, hydrate data as it arrives.
- **Save on change, not continuously.** `SavedStateHandle.set` is cheap; debounce only if you're persisting drafts to Room.
- **Profiling:** test under *Don't keep activities*; watch for `TransactionTooLargeException` in logs; measure resume time.

8.14 Testing Strategy

`SavedStateHandle` is directly constructible in tests — simulate restore by seeding it with values.

```
class CheckoutRestoreTest {
    @Test fun `restores form and step from SavedStateHandle`() = runTest {
        val saved = SavedStateHandle(
            mapOf(
                "checkout_step" to 2,
                "checkout_form" to CheckoutForm(name = "Ada", address = "1 St"),
            )
        )
        val vm = CheckoutViewModel(saved, FakeObserveCart(), FakePlaceOrder())
        assertEquals(2, vm.step.value)
        assertEquals("Ada", vm.form.value.name)         // survived "process death"
    }

    @Test fun `field change is written to SavedStateHandle`() = runTest {
        val saved = SavedStateHandle()
        val vm = CheckoutViewModel(saved, FakeObserveCart(), FakePlaceOrder())
        vm.onFieldChange { it.copy(city = "Paris") }
        assertEquals("Paris", saved.get<CheckoutForm>("checkout_form")?.city)
    }

    @Test fun `resumed submit reuses idempotency key`() = runTest {
        val saved = SavedStateHandle(mapOf("checkout_idem_key" to "k-123"))
        val placeOrder = RecordingPlaceOrder()
        val vm = CheckoutViewModel(saved, FakeObserveCart(), placeOrder)
        vm.submit()
        assertEquals("k-123", placeOrder.lastKey)        // no new key ⇒ no duplicate
    }
}
```

Also: an instrumentation test using `ActivityScenario.recreate()` (config change) plus a manual process-death pass (*Don't keep activities*) to verify end-to-end restoration; `rememberSaveable` state restoration via `StateRestorationTester`.

8.15 Production Observability

- **Metrics:** restore-after-death rate (how often users hit it), lost-input incidents (ideally 0), `TransactionTooLargeException` count, resume time.
- **Logs:** `restore{screen, restored=true/false}` (no field contents — PII), `bundle_too_large{screen, bytes}`.
- **Crash signals:** `TransactionTooLargeException` → something big in the Bundle; duplicate orders after resume → missing idempotency key.
- **Debugging:** "users lose the form" → not using `SavedStateHandle` (only ViewModel); "duplicate order after returning" → re-submitting on restore without idempotency; "slow resume" → blocking on SSOT read.
- **Privacy:** never log saved state values; they often contain user input.

8.16 System Design Interview Preparation

> ≥25 questions.

Fundamentals

1. *Config change vs process death?* ViewModel survives config change, not process death; the latter destroys all memory and recreates from saved state.
2. *What survives process death?* `SavedStateHandle` / `rememberSaveable` (Bundle) + anything in Room/DataStore/files.
3. *What's `SavedStateHandle`?* A Bundle-backed key-value map injected into ViewModels; also exposes `getStateFlow`.
4. *Why not save everything in the Bundle?* Tiny size limit (`TransactionTooLargeException`); it's serialized on the main thread.
5. *Where does durable data come from on restore?* The SSOT (Room/DataStore) or a re-fetch — not the Bundle.

Architecture

6. *What belongs in `SavedStateHandle` vs Room?* Small transient journey keys vs large/durable/relational data.
7. *How re-derive the screen after death?* Restore minimal keys, read SSOT, rebuild `UiState`.
8. *Pure-UI state across death?* `rememberSaveable` (+ custom `Saver`).

Kotlin/Android

9. *How is `SavedStateHandle` provided?* By Hilt/ `SavedStateViewModelFactory` ; `@Parcelize` for custom types.
10. *`remember` vs `rememberSaveable`?* `remember` survives recomposition only; `rememberSaveable` survives config change + process death.
11. *Custom type in `rememberSaveable`?* Provide a `Saver` .

Concurrency/correctness

12. *Duplicate side effect on resume?* Persist an idempotency key (Ch 15); re-derive, don't replay.
13. *In-flight coroutine on death?* Lost; use WorkManager for durable work (Ch 30); UI re-derives from state.
14. *Restore vs newer SSOT data conflict?* Prefer the SSOT (authoritative).

Limits

15. *Bundle size limit?* ~1 MB per Binder transaction; keep saves to bytes/KB.
16. *What triggers `TransactionTooLargeException`?* Oversized saved state (lists, bitmaps).

Security

17. *Secrets in the Bundle?* Never — unencrypted on disk; use Keystore (Ch 20).
18. *PII drafts?* Persist to encrypted Room (Ch 24), excluded from backup (Ch 37).

Scalability

19. *Multi-screen journey restoration?* A restorable-state contract + nav back stack restoration (Ch 53).
20. *Scroll position in huge lists?* Paging + saved key; `rememberLazyListState` saver.

Testing

21. *How test restoration?* Seed `SavedStateHandle` in a unit test; assert derived state.
22. *How force process death? Don't keep activities* / `adb shell am kill`.
23. *Why isn't rotation enough?* Rotation keeps the ViewModel; it doesn't exercise Bundle restore.

Production debugging

24. *Users lose form input — cause?* State only in ViewModel/composable, not `SavedStateHandle`.
25. *Duplicate order after returning — cause?* Re-submit on restore without an idempotency key.
26. `TransactionTooLargeException` — *cause?* Saving too much in the Bundle.

Design-change challenges

1. **Long draft must never be lost.** Persist to Room with a debounced save (Ch 37); keep only a draft id in the Bundle.
2. **Multi-step wizard must resume at the right step.** Save `step` + per-step form in `SavedStateHandle`.
3. **Payment interrupted by process death.** Idempotency key persisted; backend dedupe; re-derive status from the SSOT/backend.
4. **Restore scroll + selection in an infinite list.** Paging + saved anchor key + list-state saver.
5. **Restore the whole navigation journey.** Nav back-stack state restoration (Ch 53) + per-destination `SavedStateHandle`.

Mock walkthrough (abridged)

Clarify (what must survive? how big? any in-flight mutation?) → split state into Bundle (tiny keys) vs SSOT (durable) → `SavedStateHandle` + `rememberSaveable` → idempotency for in-flight → size limits/security → test via *Don't keep activities*.

8.17 Practical Exercises

1. **Beginner.** Save and restore a search query with `SavedStateHandle.getStateFlow`; verify via a seeded handle. *Eval:* query survives simulated death.
2. **Intermediate.** Restore a multi-field form + current step with `@Parcelize`; add `rememberSaveable` for an expanded section. *Eval:* full resume.
3. **Advanced.** Make an order submission safe across process death with a persisted idempotency key; prove no duplicate on resume. *Eval:* same key reused; backend dedupes.
4. **Production debugging.** Fix a screen that loses input on process death (state only in ViewModel) and one that throws `TransactionTooLargeException` (list in Bundle). *Eval:* Bundle holds ids; data from SSOT.
5. **System design challenge.** Persist a long draft to Room with debounced saves, keeping only a draft id in the Bundle; restore seamlessly. *Eval:* draft survives; Bundle tiny.

(Solutions withheld unless requested.)

8.18 Chapter Summary & Reference Sheet

Key decisions: `ViewModel` survives rotation but **not** process death; save the **minimal transient keys** in `SavedStateHandle` / `rememberSaveable` and **recover durable data from the SSOT**; keep the Bundle tiny (hard size limit); use **idempotency keys** so resumed mutations don't duplicate; **never** store secrets/PII in the Bundle; persist long drafts to Room, not the Bundle.

Concepts: process death vs config change, saved instance state, Bundle size limit, `SavedStateHandle` / `getStateFlow`, `rememberSaveable` / `Saver`, re-derive vs replay, idempotent restore.

Common mistakes: only testing rotation; state only in the ViewModel (lost on death); big objects in the Bundle (`TransactionTooLargeException`); secrets in the Bundle; re-submitting operations on restore (duplicates); persisting one-time effects (Ch 7).

Checklist:

- [ ] transient keys in `SavedStateHandle`
- [ ] durable data from SSOT
- [ ] Bundle tiny
- [ ] `@Parcelize` / `Saver` for custom types
- [ ] idempotency key for in-flight mutations
- [ ] no secrets/PII in Bundle
- [ ] long drafts → Room
- [ ] tested via *Don't keep activities*.

Key APIs: `SavedStateHandle` (`get` / `set` / `getStateFlow` / `contains`), `@Parcelize`, `rememberSaveable`, `Saver` / `mapSaver` / `listSaver`, `rememberLazyListState`, `ActivityScenario.recreate`, `StateRestorationTester`.

References:

- SavedStateHandle / ViewModel: https://developer.android.com/topic/libraries/architecture/viewmodel/viewmodel-savedstate
- Saving UI state: https://developer.android.com/topic/libraries/architecture/saving-states
- Compose state saving (`rememberSaveable`): https://developer.android.com/develop/ui/compose/state#restore-ui-state
- Processes and app lifecycle: https://developer.android.com/guide/components/activities/process-lifecycle

Chapter review (Phase 4)

- **Completeness:** all 18 sections; architecture/class/sequence/failure diagrams; `SavedStateHandle` + `rememberSaveable` + Saver + idempotent restore code; concurrency; failure matrix; 5 levels; store-selection trade-offs; security (Bundle on disk); perf/limits; seeded-handle + restoration tests; observability; 26 Q&A + 5 challenges. ✅

- **Expand later:** durable drafts (Ch 37), nav back-stack restoration (Ch 53), WorkManager continuation (Ch 30).
- **Interview coverage:** exceeds ≥25 bar. **Part I is now complete (Ch 1–8).**